

4.2 Reviewing the process of creating the Augmented Reality (AR) model prototype

□ Ways to review ▪ The effectiveness of the processes followed ▪ The effectiveness of the tools and techniques used ▪ Does the AR model prototype meet the defined purpose ▪ Lessons learnt	To include: • The importance of using the design documentation • The effective use of AR tools and techniques to create an AR model prototype • Considering lessons learnt to improve approach when planning and creating an AR model prototype • Consideration of the defined purpose of the AR model prototype
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Marking criteria

[Section 6.4](#) provides full information on how to mark the NEA units and apply the marking criteria. The marking criteria descriptor words are further explained in [Appendix B](#).

The tables below contain the marking criteria for the tasks for this unit. If a student's work does not meet any Mark Band 1 (MB1) criteria for the task, you must award zero marks for that task.

Unit R070 - Topic Area 2: Designing an AR model prototype		
MB1: 1–3 marks	MB2: 4–6 marks	MB3: 7–10 marks
Limited use of design tools and features used to plan the prototype, which are under-utilised for the creation of planning documentation	Adequate use of design tools and features used to plan the prototype, which are utilised for the creation of planning documentation	Effective use of design tools and features used to plan the prototype, which are fully utilised for the creation of planning documentation
MB1: 1–4 marks	MB2: 5–8 marks	MB3: 9–13 marks
Limited planning documentation analysing user requirements. Requirements are identified	Adequate planning documentation analysing user requirements. Requirements are described	Effective planning documentation analysing and meeting user requirements. Requirements are explained
Limited design documentation for technical development	Adequate design documentation for technical development	Effective design documentation for technical development

MB1: 1–3 marks	MB2: 4–6 marks	MB3: 7–10 marks
Limited use of tools and features which are under-utilised for the creation of the prototype Basic user interaction within model prototype	Adequate use of tools and features which are utilised for the creation of the prototype Adequate user interaction within model prototype	Effective use of tools and features which are fully utilised for the creation of the prototype Effective user interaction within model prototype
MB1: 1–4 marks	MB2: 5–8 marks	MB3: 9–13 marks
The solution allows limited information to be presented to the intended audience(s) in a limited manner The solution will contain inefficiencies that affect the quality of the information presented The solution will provide a limited user experience	The solution allows adequate information to be presented to the intended audience(s) The solution will contain inefficiencies, but these do not affect the quality of the information presented The solution provides an adequate user experience	The solution allows effective information to be presented to the intended audience(s) The solution is effective and efficient in presenting good quality and relevant information The solution provides an effective user experience

Unit R070 – Topic Area 4: Testing and reviewing

MB1: 1–2 marks	MB2: 3–4 marks	MB3: 5–7 marks
Testing is limited with minimal identification of tests used and results stated	Testing is adequate for most aspects with results documented. Any changes made due to test results are commented upon	Testing is effective for all aspects with results thoroughly and completely documented. Any changes made due to test results are explained
MB1: 1–2 marks	MB2: 3–4 marks	MB3: 5–7 marks
Basic review which states what worked well and/or what did not work well	Adequate review which describes the effectiveness of the processes followed during the designing and creation of the AR solution Adequate review which describes the effectiveness of the tools and techniques used during the designing and creation of the AR solution	Comprehensive review which explains the effectiveness of the processes followed during the designing and creation of the AR solution Comprehensive review which explains the effectiveness of the tools and techniques used during the designing and creation of the AR solution. Describes lessons learnt

Assessment guidance

During the NEA, students will be expected to demonstrate their skills and understanding through 3 to 5 tasks in the set assignment. Students are required to plan and design, create, test and review their Augmented Reality (AR) model prototype solution for the particular context. Students must complete the assignment independently therefore it is highly unlikely that the same AR model prototype solution is created by all students. Students **can use** an augmented reality (AR) software development kit (SDK), **or** alternative software to complete the assessment. For more information see [Appendix C](#). If alternative software is used students **must** create a simulated AR experience. Alternative software can include, but isn't limited to, presentation software (e.g. PowerPoint, Google Slides and Canva).

A simulated AR experience **must show what AR would look like when it is working**. This means using things that move or change, such as simple animation, clicking or tapping to make something happen, and visual effects including making objects appear closer or further away to look like they are in the same space as the real world. The simulated AR experience must include the illusion of depth.

When an AR model prototype is made using an **AR SDK**, animation may be applied through built-in functionality (e.g. via software tools). When using **alternative software**, such as presentation software students **must create these effects themselves** using tools like animations, layers, size changes, and slide transitions. In both cases, the aim is to show how virtual objects would look and behave in AR, not just to create a static set of slides.

Students **must** complete the full assignment before any evidence is marked. For Task 2, students **must** submit a video or screen capture demonstrating the AR model prototype in use. Teachers **must** assess Task 2 using only this recording. Marks based on alternative forms of evidence cannot be accepted. Work to be submitted for moderation **must** include the full recording. Moderators **must** see the recording and be confident the marks are based on that recording.

Teaching content	Assessment Guidance
1.1, 1.2 and 1.3	- Students must know and understand the different uses of Augmented Reality (AR) in the different sectors, the different types of AR and the user interactions and the devices used for AR to support their explanations when designing the AR product for the assignment context. - While this aspect is not directly assessed, students know and understand this topic area in order to be able to complete the set assignment.
3.1, 3.2, 3.3 and 3.4	- Students must only create an AR model prototype meaning it is a simple model built and not a fully working AR product. Their AR model prototype will have limited functionality to showcase how the fully working AR solution for a given context and audience might look like. Think of it as a minimal viable product – just enough to demonstrate to your client or customer what the full version will look like and how it might function. - Students can use an augmented reality (AR) software development kit (SDK), or alternative software to be able to use the tools and techniques as listed in Topic 3 above to create, test and review their AR model prototype. For more information see Appendix C. If alternative software is used students must create a simulated AR experience.
4.1	Students are required to carry out technical testing only as part of their assignment. They must not share any aspect of their assignment or prototype. Therefore, students must not carry out user testing.

Topic Area		Topic Area	
2	Designing an Augmented Reality (AR) model Prototype	1	Design tools
		5	Digital communications
4	Testing and reviewing	3	Data and testing

More information about synoptic assessment within this qualification can be found in [section 5.2 Synoptic assessment](#).

J836	OCR Level 1/Level 2 Cambridge National in IT	120*	603/7115/8
Made up of three mandatory units:			
Units R050, R060 and R070.			

*the GLH includes assessment time for each unit.

Individual unit details below:

Unit R050: IT in the digital world	
48 GLH 1 hour 30 minute written examination 70 marks (80 UMS) OCR-set and marked Calculators are not required in this exam	This question paper has two sections: - Section A – worth 15 marks. Includes closed response, multiple choice and short response questions. - Section B – worth 55 marks. Includes scenario based short, medium and extended response questions. One question will be a create style question [8 marks]. One extended response question [9 marks] will be assessed using a levels of response mark scheme.
Unit R060: Data manipulation using spreadsheets	
36 GLH OCR-set assignment 60 marks (60 UMS) Centre-assessed and OCR moderated	This set assignment contains three to five practical tasks. It should take approximately 10-12 GLH to complete.
Unit R070: Using Augmented Reality to present information	
36 GLH OCR-set assignment 60 marks (60 UMS) Centre-assessed and OCR moderated	This set assignment contains three to five practical tasks. It should take approximately 10-12 GLH to complete.

The terminal assessment rule or 'terminal rule' means that the exam must be taken in the final assessment series of the student's course. Non examined assessment (NEA) units can be submitted in the same series as the exam or an earlier series but the exam must be taken in the final series.

If a student takes the exam in a series before their work for NEA units are submitted, this is considered as a 'practice attempt and will not contribute to their final qualification grade. The student must take their exam again in their final assessment series and that exam result will be used towards the student's final qualification grade.

For more information on the terminal rule and also what it means for performance tables see [Section 7.3](#). Resitting units before certification and Retaking the qualification.

OCR-set assignments for units R060 and R070 will be available on our secure website for teachers, [Teach Cambridge](#).

5.2 Synoptic assessment

Synoptic assessment is a built-in feature of this qualification. It means that students need to use an appropriate selection of their knowledge, understanding and skills developed across the qualification in an integrated way and apply them to a key task or tasks.

This also helps students to build a holistic understanding of the subject and the connections between different elements of learning, so they can go on to apply what they learn from this qualification to new and different situations and contexts.

The externally assessed unit R050 allows students to gain underpinning knowledge and understanding

relevant to IT and the non examined assessment (NEA) units R060 and R070 draw on and strengthen this learning by letting students apply their learning in a practical, skills-based way.

It is important to be aware of the synoptic links between the units so that teaching, learning and assessment can be planned accordingly. Then students can apply their learning in ways which show they are able to make connections across the qualification when they are assessed.

5.3 Transferable skills

This qualification also allows students the opportunity to gain broad, transferable skills and experiences that can be applied as they progress into their next stages of study and life and to enhance their preparation for future employment.

Students will develop the following skills that are transferable to different real-life contexts, roles or employment:

- planning a sequence of processes. This involves identifying hardware and software resources required for the solution, as well as reviewing plans, if necessary
- solving problems by identifying and rectifying any faults in the solution
- presenting ideas and concepts digitally, identifying and choosing the right tools, text, images, audio or video, etc
- analysing information and showing/demonstrating the ability to visualise, problem-solve, and make decisions
- thinking creatively to explore and generate ideas/solutions and outcomes that are of value. As part of this process and development:
 - connect ideas and experiences in inventive ways
 - question assumptions being made
 - try out alternatives or new solutions and follow ideas through
 - adapt ideas as circumstances change.

5.4 Grading and awarding grades

All results are awarded on the following scale:

- Distinction* at Level 2 (*2)
- Distinction at Level 2 (D2)
- Merit at Level 2 (M2)
- Pass at Level 2 (P2)
- Distinction at Level 1 (D1)
- Merit at Level 1 (M1)
- Pass at Level 1 (P1).

The shortened format of the grade will show within results files and results reports. However, the full format of the grade will be on the certificates issued to students.

The boundaries for Distinction at Level 2, Pass at Level 2, and Pass at Level 1 are set judgements. Other grade boundaries are set arithmetically.

The Merit (Level 2) is set at half the distance between the Pass (Level 2) grade and the Distinction (Level 2) grade. Where the gap does not divide equally, the Merit (Level 2) boundary is set at the lower mark (For example, 45.5 would be rounded down to 45).

For the examined unit, the Distinction* (Level 2) grade is normally set at about 0.75 of the D2-M2 distance above the D2 boundary mark.

To set the Distinction (Level 1) and Merit (Level 1) boundaries, the gap between the Pass (Level 1) grade and the Pass (Level 2) grade is divided by 3, and the boundaries set equidistantly. Where this division leaves a remainder of 1, this extra mark will be added to the Distinction (Level 1) to Pass (Level 2) interval, meaning the Distinction (Level 1) boundary will be lowered by 1 mark. Where this division leaves a remainder of 2, the extra marks will be added to the Distinction (Level 1) to Pass (Level 2) interval, and the Merit (Level 1) to Distinction (Level 1) interval, meaning the Distinction (Level 1) boundary will be lowered by 1 mark, and the Merit (Level 1) boundary will be lowered by 1 mark.

For example, if Pass (Level 2) is set judgementally at 59, and Pass (Level 1) is set judgementally at 30, then Distinction (Level 1) is set at 49, and Merit (Level 1) is set at 39.

Grades are indicated on qualification certificates. However, results for students who fail to achieve the minimum grade (Pass at Level 1) will be recorded as unclassified (U or u) and **will not** be shown on certificates.

This qualification is unitised. Students can take units across different series and can resit units (see [section 7.8 Unit and qualification resits](#)). Grade boundaries are set per unit, per series, so may be set in different places for a unit in different series. When working out students' overall grades, OCR needs to be able to compare performance on the same unit in different series when different grade boundaries may have been set, and between different units. We use a Uniform Mark Scale (UMS) so this can be done.

A student's uniform mark for each unit is calculated from the student's raw mark on that unit. The raw mark boundary marks are converted to the equivalent uniform mark boundary. Marks between grade boundaries are converted on a pro rata basis.

When unit results are issued, the student's unit grade and uniform mark are given. The uniform mark is shown out of the maximum uniform mark for the unit (for example, 42/60).

The table below shows the Raw marks and UMS marks for each unit:

Marks	Exam	NEA1	NEA2
Raw marks	70	60	60
UMS	80	60	60

The uniform mark boundaries for each of the assessments do not change and are shown below:

Unit GLH	Max Unit Uniform Mark	Unit Grade							
		Distinction* at L2	Distinction at L2	Merit at L2	Pass at L2	Distinction at L1	Merit at L1	Pass at L1	U
36	60	54	48	42	36	30	24	18	0
48	80	72	64	56	48	40	32	24	0

The student's uniform mark for Unit R050 will be combined with the uniform mark for the NEA units to give a total uniform mark for the qualification. The student's overall grade will be determined by the total uniform mark. The following table shows the minimum total mark for each overall grade:

Max Uniform Mark	Qualification Grade							
	Distinction* at L2	Distinction at L2	Merit at L2	Pass at L2	Distinction at L1	Merit at L1	Pass at L1	U

5.5 Performance descriptors

Performance descriptors give a general indication of likely levels of attainment by representative students performing at boundaries: Distinction at Level 2, Pass at Level 2 and Pass at Level 1.

Performance descriptor – Distinction at Level 2

Students will be able to:

- recall, select and apply **detailed** knowledge and **thorough** understanding of IT
- present information **clearly** and **accurately**, using a **wide range** of technical language and IT terminology
- apply **relevant** knowledge, understanding and skills in a **range** of situations to plan and create IT solution and product **effectively**, testing their solutions, and working safely
- analyse and evaluate the evidence available, reviewing and adapting their methods **where appropriate**
- make **reasoned** judgements and **substantiated** conclusions
- work confidently and independently to create solution and product which reflects **thoughtful** planning, **skilled** development and **perceptive** evaluation as well as **actively demonstrating** practical skills at a **high level**.

Performance descriptor – Pass at Level 2

Students will be able to:

- recall, select and apply **sound** knowledge and understanding of IT
- present information **clearly** and with **some accuracy**, using a **range of** technical language and IT terminology
- apply knowledge, understanding and skills in a **range** of situations to plan and create IT solution and product, testing their solutions, and working safely
- review evidence available, analysing and evaluating **some** information **clearly** and making **some basic** adaptations to their methods
- make **judgements** and draw **appropriate** conclusions
- work independently to create solution and product which reflects **effective** planning, development and evaluation and an ability to demonstrate **sound** practical skills.

Performance descriptor – Pass at Level 1

Students will be able to:

- recall, select and apply knowledge and understanding of **basic** aspects of IT
- present **basic** information, using **limited** IT terminology
- apply **limited** knowledge, understanding and skills to plan and create **simple** IT solution and product, with an awareness of the need for safety
- review evidence and draw **basic** conclusions
- create solution and product which demonstrates a degree of planning, development and evaluation and **limited** practical skills.

6 Non examined assessment (NEA) units (R060-R070)

This section provides guidance on the completion of the NEA units (R060 and R070).

The NEA units are designed so that students can build a portfolio of evidence to meet the topic areas for the unit.

Assessment for this qualification must adhere to JCQ's [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#).

Please **do not** use JCQ's Instructions for Conducting Non-examination Assessments (GCE and GCSE

Specifications) – these are only relevant to A Level and GCSE specifications.

Units R060 and R070 are centre assessed and externally moderated by us.

You **must** make sure that you have read and understood all of the rules and guidance provided in this section **before** your students complete and you assess the set assignments.

If you have any queries please [contact us](#) for help and support.

6.1 Preparing for NEA unit delivery and assessment

6.1.1 Centre and teacher/assessor responsibilities

For the NEA units of this qualification we assume the teacher is the assessor.

Before you plan to get [approval](#) from us to offer this qualification you must be confident your centre can fulfil all the responsibilities described below. Once you're approved, you can offer any of our general qualifications, Cambridge Nationals, Cambridge Advanced Nationals or enrichment qualifications **without** having to seek approval for individual qualifications.

The quality of the delivery of teaching and the integrity of assessments and quality assurance is paramount. Systems must be in place so that assessments are fair, valid, reliable and authentic. One of the key factors behind valid, fair and reliable assessment is the expertise of those doing the assessment and internal quality assurance.

With this in mind, here's a summary of the responsibilities that your centre and teachers must be able to fulfil. It is the responsibility of the head of centre¹ to make sure our requirements are met:

- there are enough trained or qualified people to teach and assess the expected number of students you have in your cohorts and they will complete the **Cambridge Nationals: Essentials for the NEA** online training, prior to assessment of the set assignment. This is available on-demand and accessed through [Teach Cambridge](#) or [Cambridge OCR Train](#)
- teaching staff have the relevant level of subject knowledge and skills to deliver and assess this qualification
- teaching staff will fully cover the knowledge, understanding and skills requirements in teaching and learning activities
- necessary resources are available for teaching staff and students during teaching and assessment activities, to give students every opportunity to meet the requirements of the qualification and reach the highest grade possible
- there's a system of standardisation in place so that all assessment decisions for teacher-marked (centre assessed) assignments are consistent, fair, valid and reliable (see [internal standardisation](#) in section 6.4.3)
- there's enough time for effective teaching and learning, assessment and internal standardisation
- processes are in place to make sure that students' work is individual and confirmed as being authentic (see [Ways to authenticate work](#) in section 6.2.1)
- you must use the OCR-set assignments for students' summative assessments. You must make sure that students use the assignment that is live for the period during which they are taking their summative assessment.
- the OCR-set assignments must not be used for practice (see section 6.2, [Requirements and guidance for delivering and marking the OCR-set assignments](#)). This includes both assignments that are currently live or live assignments that have

- expired. Sample assessment material for each of the NEA units is available on Teach Cambridge and the OCR website. The sample assessment material can be used for practice purposes
- students understand what they need to do to get the highest marks possible
 - students understand what it means when we say work must be authentic and individual and they (and you) must follow any requirements we set out to make sure their work is their own
 - students know they must not reference another individual's personal details in any evidence produced for summative assessment in accordance with the Data Protection Act 2018 and the UK General Data Protection Regulations (UK GDPR). It is the student's responsibility to make sure evidence that includes another individual's personal details is anonymised
 - marks submitted to us are correct, accurately recorded and adhere to the published deadlines
 - assessment of set assignments must adhere to the JCQ [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#) and JCQ [AI Use in Assessments: Protecting the Integrity of Qualifications](#)
 - a declaration is made at the point you're submitting any work to us for assessment that confirms:
 - all assessment is conducted according to the specified regulations
 - students' work is authentic
 - marks have been transcribed accurately
 - centre records and students' work are kept according to the requirements below:
 - students' work must be kept until after their unit has been awarded and any review of results or appeals processed. We will not consider any review if the work has not been kept
 - internal standardisation and assessment records must be kept securely for a minimum of three years after the date we've issued a certificate for a qualification
 - all cases of suspected malpractice involving teachers or students must be reported (see ['Reporting suspected malpractice'](#) in section 6.3.1).

6.2 Requirements and guidance for delivering and marking the OCR-set assignments

The assignments are set by us, taken under supervised conditions, marked by the teacher and moderated by us. Assignments will be available on our secure website, Teach Cambridge.

The set assignments give an approximate time that it will take to complete all tasks. These timings are for guidance only, but should be used by you, the teacher, to give students an indication of how long to spend on each task. You can decide how the time should be allocated between each part or individual task. You are also permitted to spread the tasks across several sessions, and therefore it is permissible for evidence to be produced over several sessions.

We will replace the set assignments each year. New set assignments will be released on 1st June. You **must**:

- only download assignments from our secure website, Teach Cambridge and use the assignments for the assessment series shown on the front cover.
- have made entries for the series you are intending to submit the NEA work for.

- not share the set assignments with anyone from outside of your centre. These must only be shared with appropriate centre staff and students taking the assessments. (More information about maintaining the integrity of assessment materials is in the JCQ document [General Regulations for Approved Centres General and Vocational qualifications](#).)
- make sure students know that they must not share assessment material or their own work with others, including posting or sharing on social media. (More information is in the JCQ guidance [Information for candidates Using social media and examinations/assessments](#).)

Assessment of the set assignments must adhere to the JCQ [Instructions for conducting non-examination assessments \(Vocational and Technical Qualifications\)](#). [Appendix A](#) of this specification gives guidance for creating electronic evidence for the NEA units. Please read Appendix A along with the unit content and marking criteria grids as it might help you plan your delivery of the units.

The rest of this section deals with how we expect you to manage the delivery and marking of the set assignments, so that assessment is valid and reliable. Please note that failing to meet these requirements may be deemed to be malpractice.

Here is a summary of what we need you to do.

You **must**:

- have covered the knowledge, understanding and skills with your students and be sure they are ready for assessment before you start the summative assessment
- use the correct live OCR-set assignment for summative assessment (the dates for which set assignments are live are shown on the front cover). These assignments are only available on **Teach Cambridge**.
- give students the [Student Guidance](#) document before they start the assessment
- make sure students are clear about the tasks they must complete and the criteria they are expected to meet. You can:
 - explain the task
 - provide a copy of the marking criteria to students
- allow students a reasonable amount of time to complete the assignments and be fair and consistent to all students. The time you allow should be in line with the estimated time we think it should take which is stated in the OCR-set assignments. Within that time students can work on the tasks under teacher supervision any time until the date the centre collects the work for centre assessment
- tell the students the resources and sources of assets that they can use in the assignment before undertaking the assessment tasks
- monitor students' progress to make sure work is capable of being assessed against the marking criteria, on track for being completed in good time and is the **student's own** work:
 - NEA work **must** be completed during normal curriculum time and supervised and marked by the teacher/assessor
 - work must be carried out with enough supervision to make sure that the work submitted can be confidently authenticated as the student's own work. You must also be familiar with the requirements of the JCQ document [AI Use in Assessments: Protecting the Integrity of Qualifications](#)

- if you provide any material to prepare students for the set assignment, you must adhere to the rules on using referencing and on acceptable levels of guidance to students set out within the Plagiarism and Feedback sections (see 6.2.2 [Plagiarism](#) and 6.3 [Feedback](#))
- students must produce their work independently (see 6.2.1 and 6.3 on [Ways to authenticate work](#) and [Feedback](#))
- you must make sure students are aware of the requirement to keep their work secure, not share with other students and keep their passwords secure
- allow students to take the initiative to improve any element of their work as they work through the assignment
- use the marking criteria to mark students' work

Before submitting a final mark to us, you can mark students' completed assignment work and allow them to repeat any element of the assignment, reworking their original evidence, provided the assignment is still live for use. We call this a reattempt. Students must have completed the whole assignment before you mark their work. Any feedback given to students on the marked work **must**:

- be factual: telling the student what you have observed, not what to do to improve their work.
- be recorded and available to the moderator (see [section 6.3 Feedback](#)).

You **must not**:

- change any aspect of the OCR-set assignments (scenarios or tasks)
- accept multiple reattempts of work where small changes have been made in response to feedback. Marking and feedback must not be an iterative process
- allow teachers or students to add, amend or remove any work after students have submitted work for moderation. This will constitute malpractice
- give detailed advice and suggestions to individuals or the whole class on how work may be improved to meet the marking criteria
- practise the OCR-set assignment tasks with the students. We provide Sample Assignments for you to use for practice purposes
- create or use practice assignments and practice data which are similar in nature to those set by us
- use past OCR-set assignments, or amend past set assignments, for practice purposes